



## Technology Data Sheet (TDS)

### TINMORRY PETG-CF

Version 1.0

#### • Basic Introduction

TINMORRY PETG-CF is a carbon fiber reinforced PETG composite. Its formula has been specially designed, tested and optimized to provide excellent processing stability, print stability and comprehensive mechanical properties. It retains the excellent toughness, water resistance, surface gloss and other properties of generic PETG, and also has higher strength, stiffness and dimensional stability owing to the addition of carbon fiber. And the CF also brings a special texture to the prints and cover the layer lines, making their surface more delicate and smooth. Besides, being compatible with most FDM printers on the market (hardened steel nozzles are recommended), available for various colors and cost-efficient, TINMORRY PETG-CF is really an ideal and worthwhile 3D printing filament.

#### • Specifications

| Subjects            | Data                                                               |
|---------------------|--------------------------------------------------------------------|
| Diameter            | 1.75 ± 0.03 mm                                                     |
| Net filament weight | 1000 g                                                             |
| Length              | 315 ± 5 m                                                          |
| Spool material      | ABS ( (Temperature resistance: 80 °C)                              |
| Spool size          | External Diameter: 198 mm; Inner Diameter: 56 mm;<br>Width: 65 mm; |

## • Recommended Printing Settings

| Subjects                          | Data                                                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drying settings before printing   | $65 \pm 5\text{ }^{\circ}\text{C}$ , $\geq 8\text{ h}$<br>(Blast drying oven, or filament drying box)                                                           |
| Printing temperature and humidity | $\leq 65\text{ }^{\circ}\text{C}$ , $\leq 25\%\text{ RH}$<br>(Put in a working filament drying box, or put in an air-tight container and sealed with desiccant) |
| Storage temperature and humidity  | $\leq 30\text{ }^{\circ}\text{C}$ , $\leq 20\%\text{ RH}$ (Sealed with desiccant)                                                                               |
| Compatible support material       | Itself or support filament for PETG                                                                                                                             |
| Compatible printer type           | Open-frame, enclosed-frame                                                                                                                                      |
| Compatible nozzle material        | Hardened Steel                                                                                                                                                  |
| Compatible nozzle size            | 0.4 (recommended), 0.6, 0.8 mm                                                                                                                                  |
| Compatible plate type             | Textured PEI Plate or other plate                                                                                                                               |
| Plate surface preparation         | Glue                                                                                                                                                            |
| Nozzle temperature                | $240 - 270\text{ }^{\circ}\text{C}$                                                                                                                             |
| Bed temperature                   | $75 - 90\text{ }^{\circ}\text{C}$<br>(depending on the printer and plate types)                                                                                 |
| Printing speed                    | $< 250\text{ mm/s}$                                                                                                                                             |
| Chamber temperature               | $< 60\text{ }^{\circ}\text{C}$                                                                                                                                  |

## • Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY PETG-CF have been tested and are shown in the tables below.

### • Physical Properties

| Subjects                        | Testing Methods                         | Data                           |
|---------------------------------|-----------------------------------------|--------------------------------|
| Density                         | ISO 1183                                | $1.32\text{ g/cm}^3$           |
| Saturated water absorption rate | $25\text{ }^{\circ}\text{C}$ , 55% RH   | 0.38%                          |
| Melt index                      | $210\text{ }^{\circ}\text{C}$ , 2.16 kg | $14.1 \pm 1.6\text{ g/10 min}$ |
| Glass transition temperature    | DSC, $10\text{ }^{\circ}\text{C/min}$   | $70\text{ }^{\circ}\text{C}$   |
| Crystallization temperature     | DSC, $10\text{ }^{\circ}\text{C/min}$   | N / A                          |
| Melting temperature             | DSC, $10\text{ }^{\circ}\text{C/min}$   | $221\text{ }^{\circ}\text{C}$  |
| Vicar softening temperature     | ISO 306, GB/T 1633                      | $74\text{ }^{\circ}\text{C}$   |
| Heat deflection temperature 1   | ISO 75, 1.82 MPa                        | $67\text{ }^{\circ}\text{C}$   |
| Heat deflection temperature 2   | ISO 75, 0.45 MPa                        | $72\text{ }^{\circ}\text{C}$   |

### • Mechanical Properties

| Properties            | Testing Methods    | Data                  |
|-----------------------|--------------------|-----------------------|
| Tensile strength (XY) | ISO 527, GB/T 1040 | $52 \pm 4\text{ MPa}$ |
| Tensile strength (Z)  | ISO 527, GB/T 1040 | $37 \pm 3\text{ MPa}$ |

|                               |                    |                              |
|-------------------------------|--------------------|------------------------------|
| Young's modulus (XY)          | ISO 527, GB/T 1040 | 2430 ± 140 MPa               |
| Young's modulus (Z)           | ISO 527, GB/T 1040 | 1620 ± 110 MPa               |
| Breaking elongation rate (XY) | ISO 527, GB/T 1040 | 9.3% ± 1.6%                  |
| Breaking elongation rate (Z)  | ISO 527, GB/T 1040 | 6.8% ± 1.7%                  |
| Bending strength (XY)         | ISO 178, GB/T 9341 | 86 ± 4 MPa                   |
| Bending strength (Z)          | ISO 178, GB/T 9341 | 61 ± 4 MPa                   |
| Bending modulus (XY)          | ISO 178, GB/T 9341 | 3070 ± 180 MPa               |
| Bending modulus (Z)           | ISO 178, GB/T 9341 | 2240 ± 150 MPa               |
| Impact strength (XY)          | ISO 179, GB/T 1043 | 59.3 ± 4.5 kJ/m <sup>2</sup> |
| Impact strength (Z)           | ISO 179, GB/T 1043 | 8.2 ± 1.7 kJ/m <sup>2</sup>  |

#### • Chemical and Other Properties

| Subjects                      | Data                                                              |
|-------------------------------|-------------------------------------------------------------------|
| Component                     | PETG, carbon fiber                                                |
| Skin irritation               | May cause allergy in some people, and wearing gloves is suggested |
| Resistance to water           | Yes                                                               |
| Resistance to oil and grease  | Resistant to most kinds of daily oil and grease                   |
| Resistance to organic solvent | Not resistant to some organic solvents                            |
| Resistance to weak acid       | Yes                                                               |
| Resistance to strong acid     | Not resistant                                                     |
| Resistance to weak alkali     | Yes                                                               |
| Resistance to strong alkali   | Not resistant                                                     |
| Flammability                  | Yes                                                               |
| Odor during printing          | Odorless                                                          |
| Odor during combustion        | Pungent                                                           |

#### • Specimen Info

| Subjects           | Data                            |
|--------------------|---------------------------------|
| Size               | Shown at the following pictures |
| Nozzle temperature | 255 °C                          |
| Bed temperature    | 70 °C                           |
| Printing speed     | 150 mm/s                        |
| Infill density     | 100%                            |
| Infill pattern     | Concentric                      |

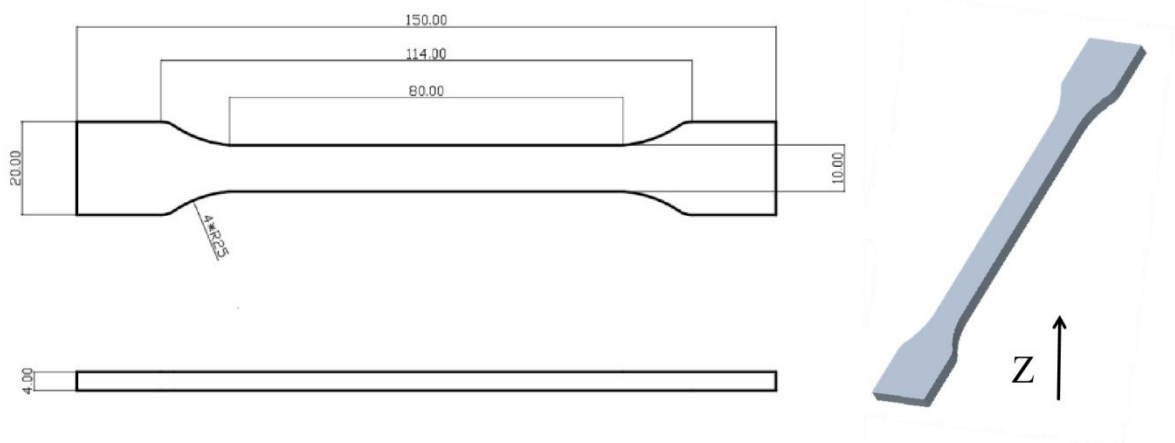
#### Statements:

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were annealed at 70 °C for 6 hours and then placed in an indoor environment at about 25 °C for 12 hours. Please note that different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z-direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY

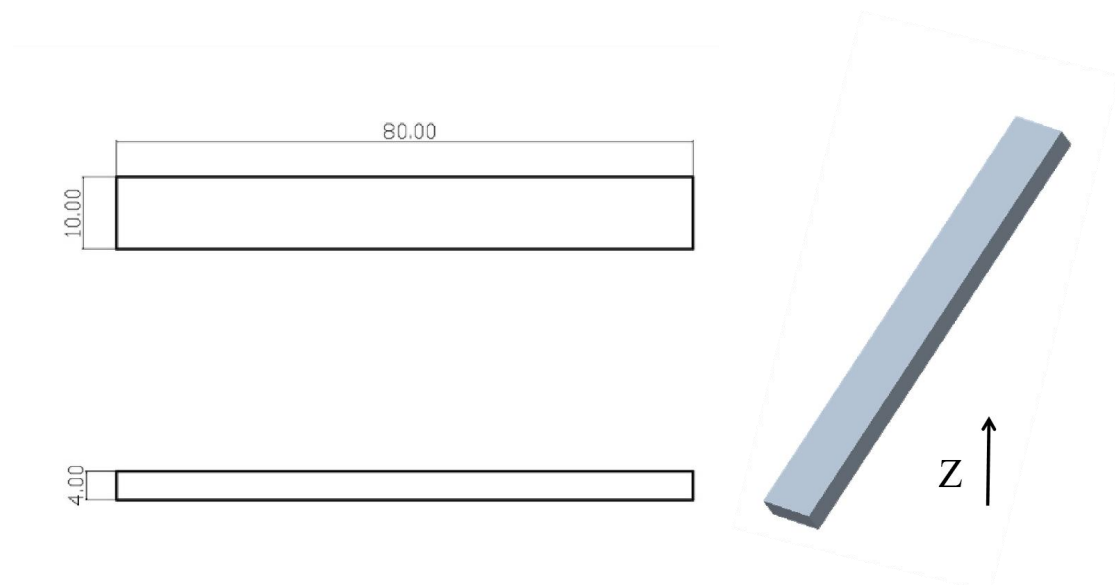
filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 25%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.

3. After annealing, the strength, stiffness, toughness and heat resistance of TINMORRY PETG-CF prints can only be increased slightly, but some prints may undergo shrinkage or deformation after the annealing process. Note that the annealing effect is also affected by the size, structure, wall thickness, and infill density of the prints themselves. If you want to anneal them, it is recommended to use a blast drying oven which has uniform temperature, and it is recommended to first anneal at about 60 °C for about 2 hours, then slowly increase the temperature to about 70 °C and keep it for about 2 to 4 hours; if the prints are significantly deformed halfway, it should be stopped immediately and a lower temperature should be used instead.

## 1. Tensile Test



## 2. Bending Test and Impact Test



## • **Disclaimer**

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them, or the usage scenarios of them.